

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in polymer and thin-film materials characterization, failure analysis, and process-property optimization (FT-IR, UV-Vis-NIR, TGA/DSC, QCM-D, GPC/SEC, electrochemistry). Currently leading DOE-guided bio-based polymer process development and translating material-quality data into reproducible scale-up specifications.

CORE COMPETENCIES

Polymer Characterization

Root Cause Failure Analysis

Thermal Analysis

DOE/SPC

Process Scale-Up

Materials Selection

Technical Documentation

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided bio-based polymer process development, converting material-quality data into reproducible process windows, scale-up specifications, and SOP-ready controls.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Assessed reliability risks in functional electrochromic materials by identifying degradation, delamination, and redox-instability failure modes across 15+ material variants using spectroscopy, electrochemistry, thermal analysis, and MS.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded thin-film coating process development, linking CVD/surface preparation to morphology, interface quality, optical performance, and process reproducibility.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Developed and optimized CVD polymer thin-film deposition with DOE strategies and in-situ QCM-D monitoring to define growth kinetics and reproducible process controls.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, and safety systems that improved instrument uptime by 30%.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Electrochromic Material Reliability Study Chemistry of Materials 2025

Mapped structure, purity, stability, and failure behavior across functional organic materials using orthogonal characterization and stress-test logic.

FT-IR, UV-Vis-NIR, CV, TGA/DSC, NMR, GPC, MS

AFOSR Defense Thin-Film Coatings DoD Funded

Established process-property relationships for CVD/surface-preparation workflows and reproducible high-performance coating conditions.

CVD, QCM-D, surface prep, thin films

Bio-Based Polymer Scale-Up Process Current

Defined DOE-guided process windows and material-quality specifications for reproducible polymer scale-up.

DOE, SPC, polymer characterization

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials

Advisor: Prof. Seth Marder

(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Reliability & Failure Analysis: Failure mode identification, root cause analysis, material degradation, stress-test design, design and process mitigation, FMEA principles

Characterization: FT-IR, UV-Vis-NIR, ellipsometry, GPC/SEC, LC-MS, GC-MS, NMR, QCM-D, TGA, DSC, cyclic voltammetry

Statistical: DOE, SPC, JMP (basic), Python (basic), Excel, statistical data analysis

Materials: Polymers, biopolymers, organic/organometallic materials, thin films, CVD coatings, surface functionalization

Process: Process development, scale-up specifications, SOPs, technical reports, reproducibility studies

JOURNAL PUBLICATIONS

- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.
- Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.
- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)